

Optoelectronic Response of WBG/UWBG Photoconductive Switches Under CW and Pulsed Laser Excitation



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WBG and UWBG semiconductors offer potential for faster photoconductive switches (PCSS) and detectors that can operate under high voltages and harsher environments. PCSS were fabricated using 4 different material stacks, and 4 different wavelengths of CW lasers were used to characterize each material's photoresponsivity. A 355 nm pulsed laser then measured the rise and fall times of the pulsed reaction. The PCSS with the thin GaN:Mn had the greatest photoresponse overall, while the thicker GaN:Mn and AlN showed faster response times.

INTRODUCTION

- Wide and Ultrawide bandgap (WBG, UWBG) semiconductors are at the forefront of new electronic devices, offering an alternative to silicon.
- Our goal is to use these devices to develop an X-ray detector. In the absence of radiation, the device will act like an insulator, allowing only a small dark current to flow. However, in the presence of even a small amount of radiation, the device will become conductive and generate large amounts of current. Thus, we would be able to detect any X-rays by monitoring the output signal from the device.

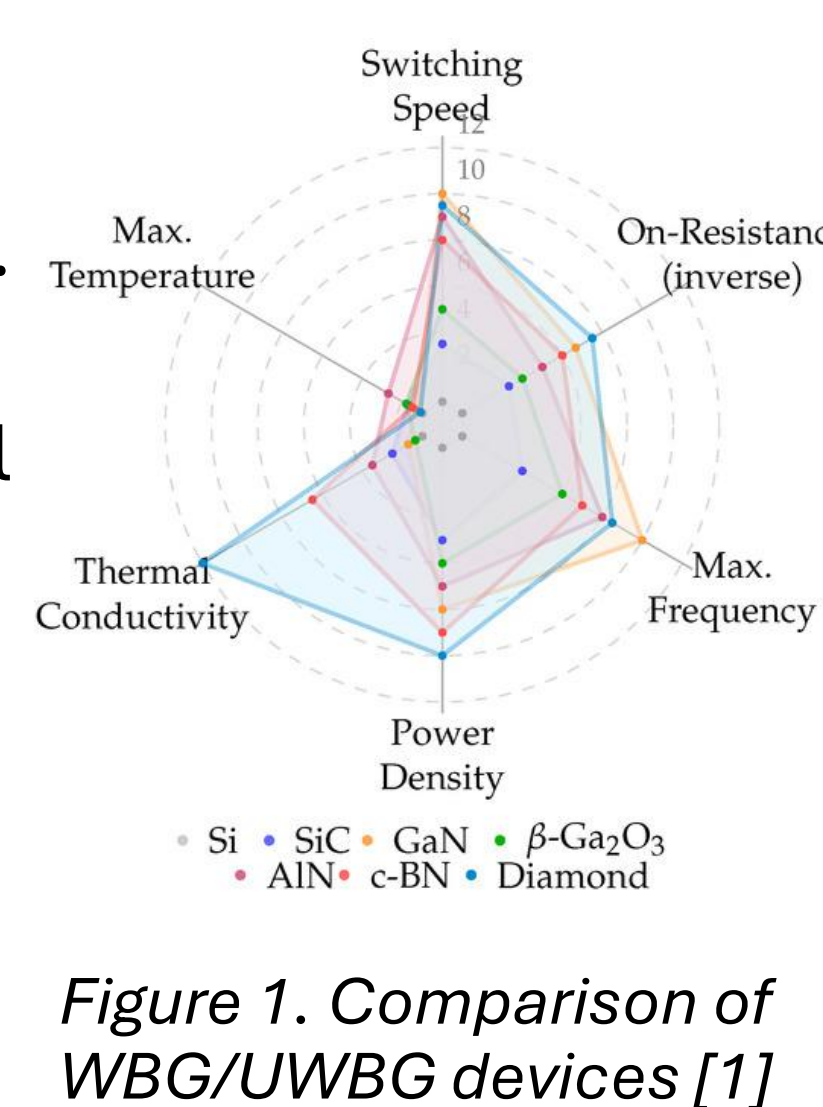


Figure 1. Comparison of WBG/UWBG devices [1]

METHODS

4 different types of PCSS were fabricated with different semiconducting materials, as seen in Table 1.

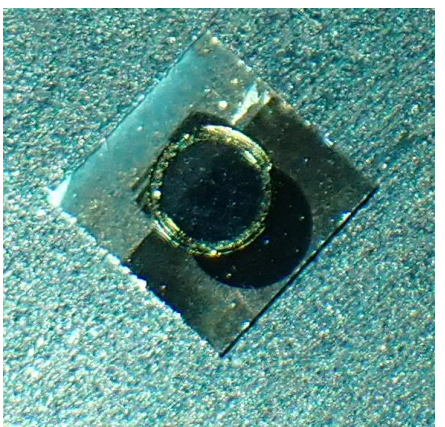
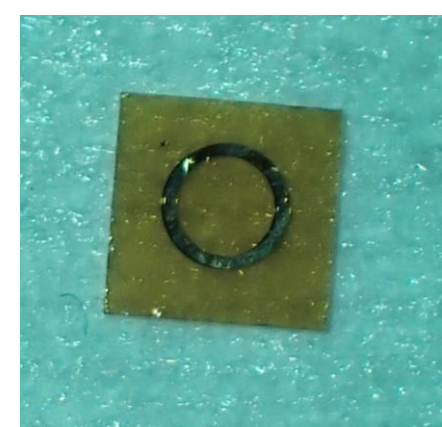
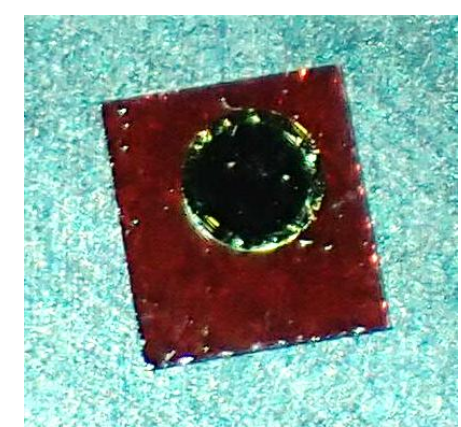
AlN:C	Gallates	Thin GaN:Mn	Thick GaN:Mn
100 μm	2.5 mm	100 μm	500 μm
			

Table 1. Summary of device names, thicknesses, and corresponding images

- A probe station equipped with a Keithley 4200 parameter analyzer was used to capture the dark and photocurrent of the devices using 4 different CW lasers (375 nm, 405 nm, 447 nm, 532 nm).
- A 20kHz, 355 nm pulsed laser setup was then used to measure the transient behavior of these devices. The rise and fall time of the signal was captured using an oscilloscope.

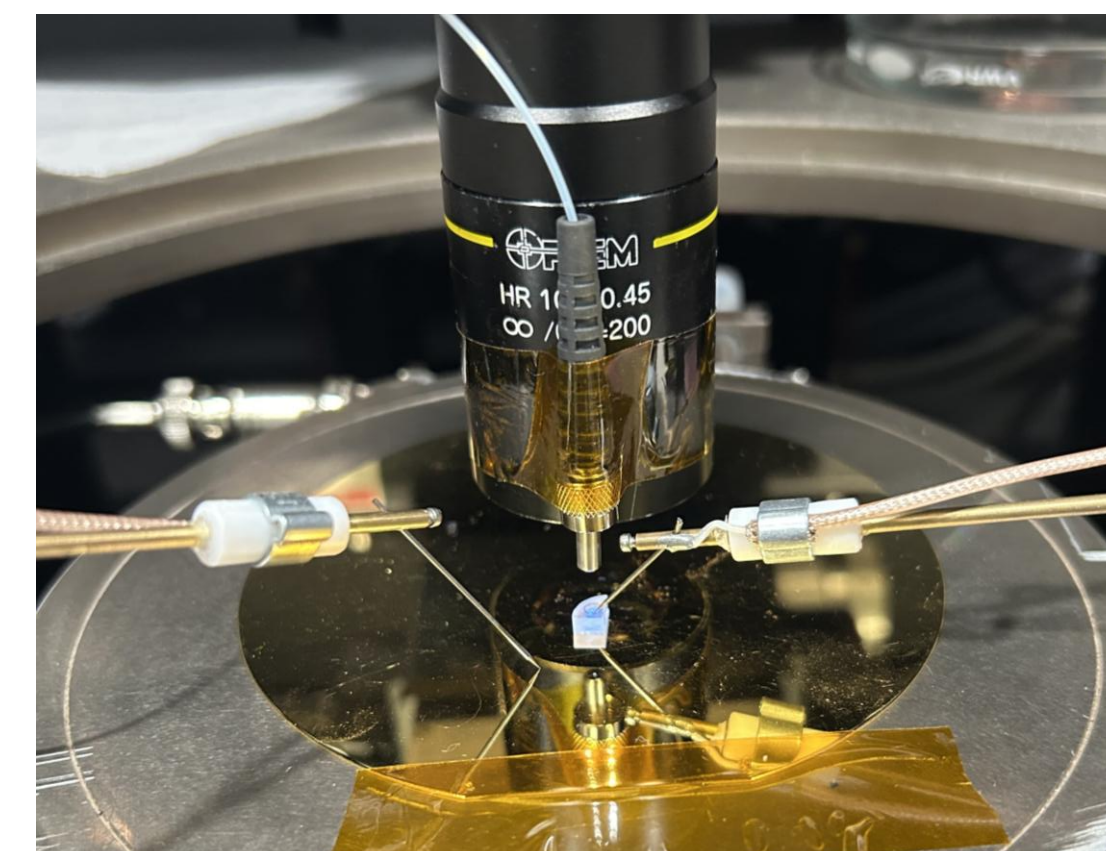


Figure 2. Probe station set up with laser diode

REFERENCES

- [1] G. Galioto, G. Vitale, A. Sferlazza, G. Lullo, and G. C. Giaconia, 'Wide and Ultrawide Bandgap Power Semiconductors: A Comprehensive System-Level Review', Electronics, vol. 15, no. 4, 2026.
- [2] Bikramjit Chatterjee, Soroush Ghandiparsi, Miranda S. Gottlieb, Erin Clark, Kathy Jackson, Clint D. Frye, Ryan D. Muir, Brandon W. Buckley, Colin Harthcock, Joel B. Varley, Sara E. Harrison, Qinghui Shao, Lars F. Voss; Optically Addressable Light Valve Based on a GaN:Mn Photoconductor. ACS Appl. Eng. Mater. 28 February 2025; 3 (2): 325–336. <https://doi.org/10.1021/acsanm.4c00664>

RESULTS

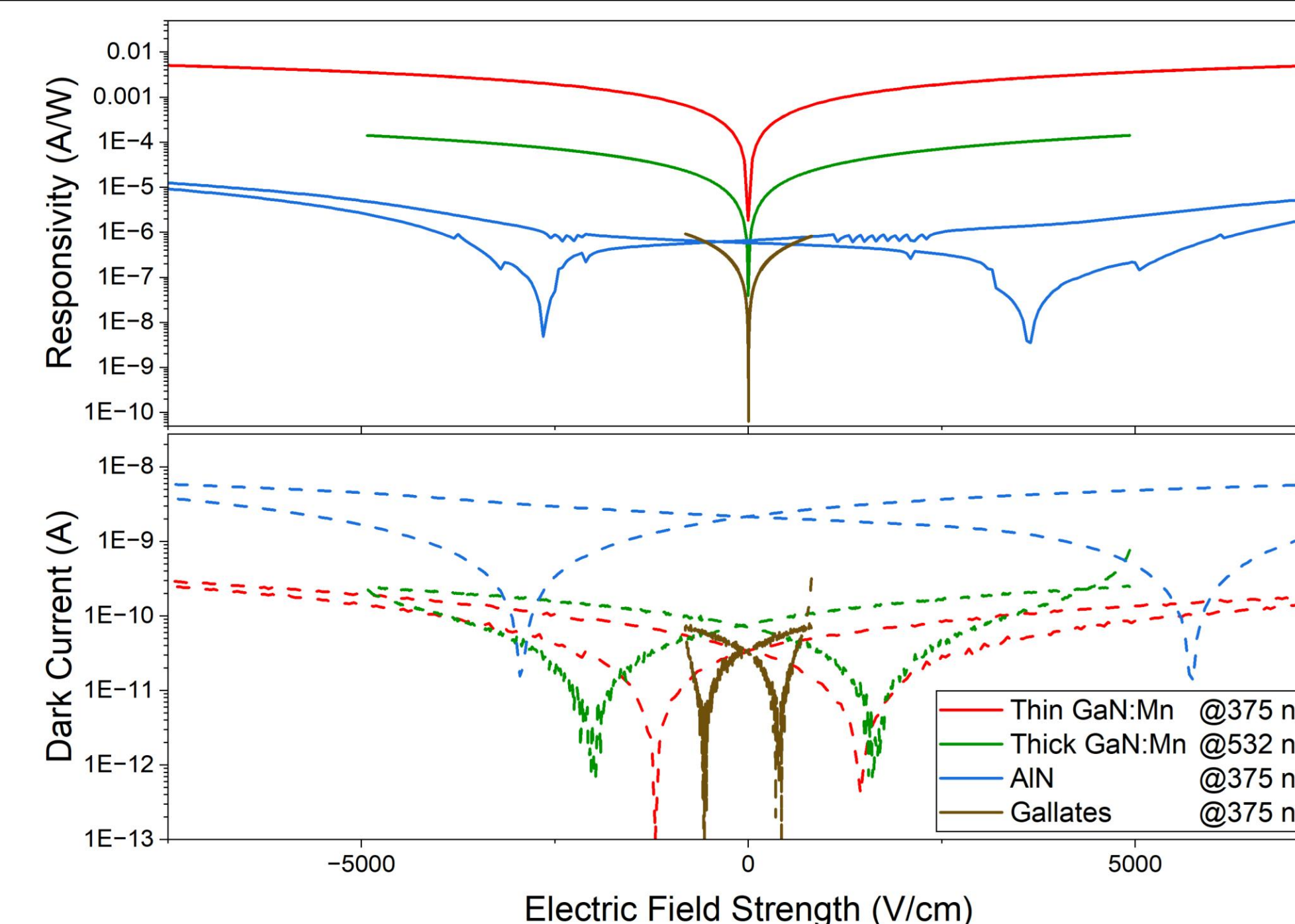


Figure 3. Responsivity and dark currents for all four devices in the best performing wavelength

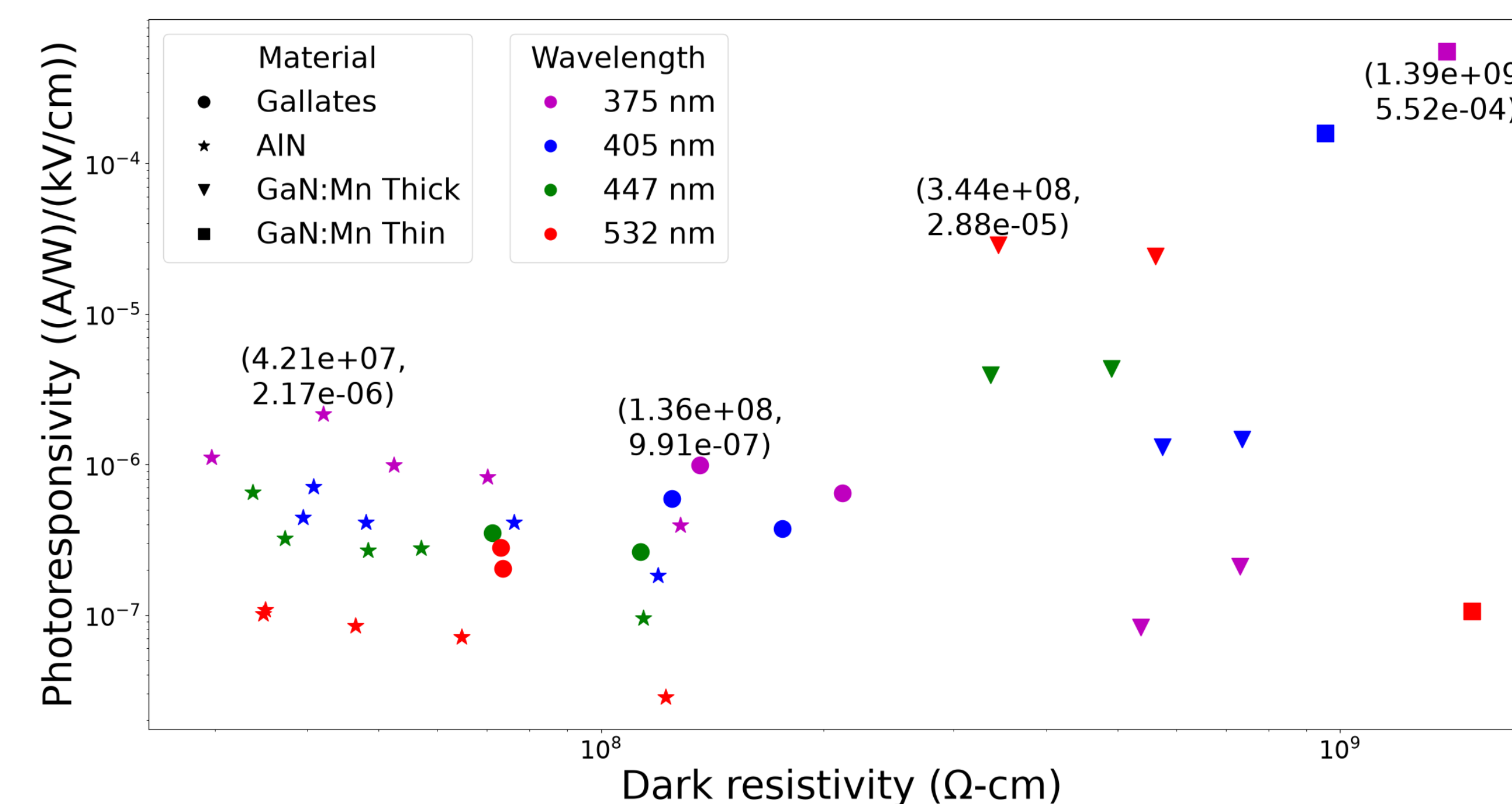


Figure 4. Dark resistivity vs Photoresponsivity plot for all devices

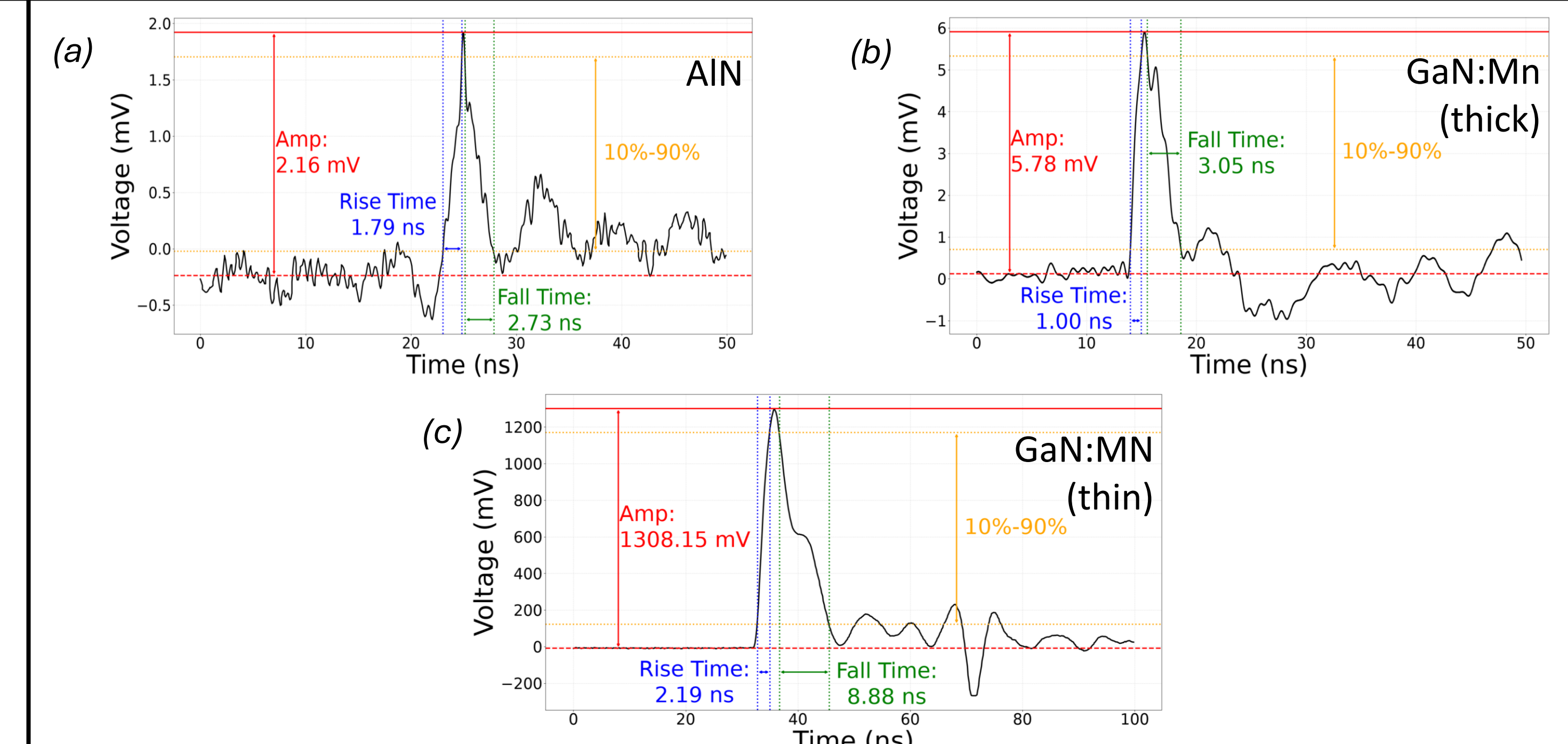


Figure 5. Timed response of the (a) AlN, (b) thick GaN:Mn and (c) thin GaN:Mn devices

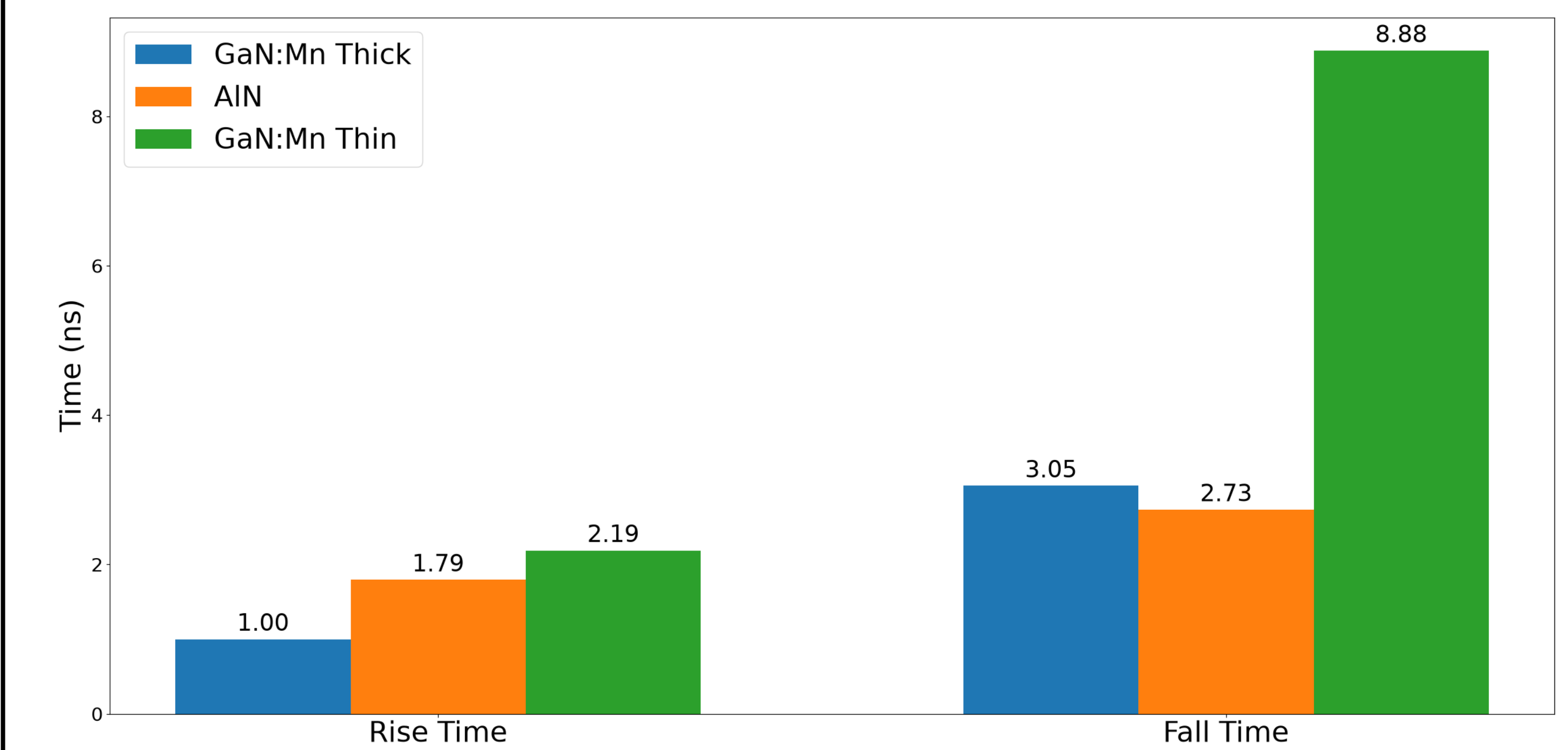


Figure 6. Comparison of rise and fall times between PCSSs

CONCLUSIONS

- The thin GaN:Mn device had the greatest photoresponse overall, with a maximum photoresponsivity of 5.52×10^{-4} (A/W)/(kV/cm) (Figure 4) at the 375 nm wavelength, which was around an order of magnitude greater than the next best performing device (Figure 3). The thin GaN:Mn, AlN and Gallates increased in photoresponsivity (Figure 4) with a decrease in wavelength, corresponding to the increase in photon energy the laser provides. However, the thick GaN:Mn device exhibited an unusual behavior of increasing in photoresponsivity with an increase in wavelength (or decrease in photon energy). This may be explained by the thickness of the GaN:Mn layer, where the material may be absorbing most of the photons before it can penetrate deeper.
- The thick GaN:Mn and the AlN devices exhibit some potential in the pulsed laser excitations. These devices have a fast rise time of 1 ns and 1.79 ns respectively, which is almost twice as fast as the rise time of the thin GaN:Mn at 2.19ns. While the fall time for the GaN:Mn thin device is approximately 4x its rise time, this interval is vastly shorter in the thick GaN:Mn (3.05x) and in AlN (1.53x), showing greater potential in faster switching speeds (Figure 5, 6).
- Because of the thickness of the thick GaN:Mn device, a laser of longer wavelength is needed for maximum photoresponse. Further testing is required to identify the optimum wavelength, and a pulsed laser test with this optimal wavelength may yield stronger output signals, while maintaining its fast rise and fall times. Similar testing can be done with the AlN devices but with lasers of shorter wavelengths.